VERY AMPLENESS OF THE SQUARE OF A GLOBALLY GENERATED INVERTIBLE MODULE WITH VANISHING H^1

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-curves/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof twice delegates to the proof of the registered sibling \label{lemma-criterion-very-ample} (source lines 1636-1637: 'all of the statements given in the proof of Lemma criterion-very-ample hold with this s'; lines 1650-1652: 'Arguing as in the proof of Lemma criterion-very-ample we find that S^{(2)} is generated in degree 1'), and ends with 'Some details omitted.' (line 1654); the essential skeleton - the descent reduction, the regular section s via lemma-automatic, the extra generator t giving surjectivity of mu_n, and the generation of S in degrees 0, 1, 2 - is present in the proof; the delegated statements are available in the registered sibling card.

Editorial note. The label lemma-criterion-very-ample-bis names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The morphism defined by a basis of global sections. Constructions, Section \label{constructions-section-projective-space}. Proof lines 1615-1617: global generation of L^{\otimes 2} gives the morphism phi to P^n_k associated to a chosen basis s_0, ..., s_n of H^0(X, L^{\otimes 2}).
- Descending the closed immersion property. Descent, Lemma \label{descent-lemma-descending-property-closed-immersion}. Proof lines 1617-1621: to check that the morphism is a closed immersion one may replace k by its algebraic closure, so assume k algebraically closed.
- Automatic Cohen-Macaulay property of the curve. Same chapter, \label{lemma-automatic}. Proof lines 1629-1633: X is Cohen-Macaulay, hence the invertible module L has no embedded associated points and the section s nonvanishing at all generic points is a regular section.
- The sibling very ampleness criterion. Same chapter, \label{lemma-criterion-very-ample}. Proof lines 1636-1637 and 1650-1652: all statements in the proof of this registered sibling hold with the chosen s, and the final generation-in-degree-1 argument for S^{(2)} is the same as in its proof.

Source reference. Original section 01ND: sources/stacks-chow/tags.txt, source lines 2154–2154. Prerequisite pointer invoked at proof line(s) 1614; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02L6: sources/stacks-chow/tags.txt, source lines 3397–3397. Prerequisite pointer invoked at proof line(s) 1617; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BY5: sources/stacks-curves/pinned-original-source.tex, source lines 1277–1277. Prerequisite pointer invoked at proof line(s) 1628; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E8V: sources/stacks-curves/pinned-original-source.tex, source lines 1506–1506. Prerequisite pointer invoked at proof line(s) 1633; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let X be a proper scheme over k having dimension 1 and $H^0(X, \mathcal{O}_X) = k$. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Assume*

(1) *$\mathcal{L}$ is globally generated,*
(2) *$H^1(X, \mathcal{L}) = 0$, and*
(3) *$\mathcal{L}$ is ample.*

Then $\mathcal{L}^{\otimes 2}$ is very ample on X over k.

Proof. Choose basis $s_0, \ldots, s_n$ of $H^0(X, \mathcal{L}^{\otimes 2})$ over k. By property (1) we see that $\mathcal{L}^{\otimes 2}$ is globally generated and we get a morphism

$$\varphi_{\mathcal{L}^{\otimes 2}, (s_0, \ldots, s_n)} : X \longrightarrow \mathbf{P}^n_k$$

See Constructions, Section 01ND. The lemma asserts that this morphism is a closed immersion. To check this we may replace k by its algebraic closure, see Descent, Lemma 02L6. Thus we may assume k is algebraically closed.

Assume k is algebraically closed. For each generic point $\eta_i \in X$ let $V_i \subset H^0(X, \mathcal{L})$ be the k-subvector space of sections vanishing at η_i. Since $\mathcal{L}$ is globally generated, we see that $V_i \neq H^0(X, \mathcal{L})$. Since X has only a finite number of irreducible components and k is infinite, we can find $s \in H^0(X, \mathcal{L})$ nonvanishing at η_i for all i. Then s is a regular section of $\mathcal{L}$ (because X is Cohen-Macaulay by Lemma 0BY5 and hence $\mathcal{L}$ has no embedded associated points).

In particular, all of the statements given in the proof of Lemma 0E8V hold with this s. Moreover, as $\mathcal{L}$ is globally generated, we can find a global section $t \in H^0(X, \mathcal{L})$ such that $t|_Z$ is nonvanishing (argue as above using the finite number of points of Z). Then in the proof of Lemma 0E8V we can use t to see that additionally the multiplication map

$$\mu_n : H^0(X, \mathcal{L}) \otimes_k H^0(X, \mathcal{L}^{\otimes 2}) \longrightarrow H^0(X, \mathcal{L}^{\otimes 3})$$

is surjective. Thus

$$S = \bigoplus\nolimits_{n \geq 0} H^0(X, \mathcal{L}^{\otimes n})$$

is generated in degrees $0, 1, 2$ over k. Arguing as in the proof of Lemma 0E8V we find that $S^{(2)} = \bigoplus_n S_{2n}$ is generated in degree 1. This means that $\mathcal{L}^{\otimes 2}$ is very ample as desired. Some details omitted. □